CompIRT: Pyramidal and Compositional Item Response Theory for Knowledge Competitions

Anonymous ACL submission

Abstract

Predicting team outcomes from individual measurements is a basic problem in psychometrics, sports analytics, organizational science, and multi-agent system design, yet no existing model connects individual assessment to team prediction through shared latent parameters. We introduce CompIRT, a probabilistic model that estimates individual skills from solo performance and composes them to predict team outcomes on collaborative tasks. *Pyramidal IRT* extends multidimensional IRT with position-dependent difficulty and a competing-risks survival model for quiz bowl tossup questions, where clues are revealed from obscure to obvious. *Compositional IRT* predicts team bonus outcomes by aggregating individual latent skills through a smooth-maximum operator and learnable pairwise synergy profiles that generalize to novel team compositions. We evaluate on nine collegiate and open-format tournaments spanning 2023–2025, using temporal train/test splits, a four-level compositional generalization test on recombined teams, and a within-tournament adaptation analysis that measures how quickly the model updates to new players. CompIRT yields interpretable skill profiles, calibrated difficulty curves, and quantified synergy effects, formalizing *individual-to-team skill prediction* as a task whose framework extends to any setting where heterogeneous agents—human or artificial—are evaluated individually and composed into teams.

1 Introduction

JBG: Proposed start text:
Human teams are common—course projects, competitive teams, collaborations—as are increasingly computer teams. But when a team completes a task successfully (or fails) who should get the credit (or blame)? This is an issue for not just classroom evaluation (grades) but also for AI evaluation (leaderboards), as agents work together with heterogeneous strengths into a collective system.

Predicting team outcomes from individual measurements is a recurring problem: medical teams diagnose from individual clinician expertise, engineering teams deliver from complementary skills, exam groups answer from pooled knowledge, and LLM ensembles combine agents with heterogeneous strengths into a collective system (Jiang et al., 2023; Wang et al., 2024). In each case, the group outcome depends not only on what each member knows, but on how their abilities compose. Item Response Theory (IRT) (van der Linden and Hambleton, 2013; Embretson and Reise, 2000), the standard framework for latent ability estimation and recently adopted for NLP evaluation (Lalor et al., 2024; Rodriguez et al., 2021), measures individuals—it has no mechanism for composing individual abilities into a team-level prediction.

JBG: Item response theory needs to be at the start of a paragraph. It's both important and likely not something people are familiar with.
Item response theory has been used to evaluate individual humans in standardized tests and computers on leaderboards, but it has not been extended to team settings. While classical IRT computes the skill of individual agents in isolation as they solve problems with estimated difficulty, this paper extends item response theory (Section ?reference? ??) to team settings with questions designed to discriminate individual skill and team collaboration.
We draw our inspiration from human question answering—a format called quiz bowl (Section ?reference? ??). This format has two components that make it well suited for this task: it has a competitive component useful for ranking agents with full visibility (**tossups**) and a cooperative component that allows us to test our measurement of individual skills (**bonuses**).

QB offers a controlled setting for studying this gap. A single QB game elicits *two* complementary assessment modes under controlled conditions. **Tossups** measure individuals: a question is read aloud one clue at a time, and any player may buzz in to answer. **Bonuses** measure teams: after a correct tossup, the team collaborates on a three-part question where only the team's collective answer counts. The same players are assessed individually and collectively within the same game, creating a direct link between individual skill estimation and team outcome prediction.

JBG: Too much detail in the introduction While this format is useful for discriminating agent ability, it requires additions to the IRT model. First, tossups break the assumption that each question has a single, fixed difficulty. To better discriminate agents, tossupos are designed to be interrupted, so we have signal not just which questions were difficult or easy but which *words* decrease the difficulty the most. Second, bonuses break the assumption that a single agent answers a single question. All agents on a team provide input on what they think the answer should be and the team comes to a consensus response.

Tossups also challenge a core assumption of classical IRT: that each item has a single, fixed difficulty. QB tossups are *pyramidal*—clues are arranged from obscure to obvious (Eltinge, 2013)—so effective difficulty *decreases* as the question is read. A player who buzzes after the first sentence faces a fundamentally harder item than one who buzzes at the giveaway clue. Modeling this requires an IRT extension where difficulty is a function of revelation position.

Three limitations of existing work motivate our approach:

1. Static item difficulty. Classical IRT and its recent NLP applications assign one difficulty parameter per item (Lalor et al., 2019; Vania et al., 2021). Response-time IRT (van der Linden, 2007) jointly models accuracy and speed but treats response time as a separate latent trait, not as a consequence of within-item difficulty variation. Neither framework handles pyramidal questions where difficulty depends on how much of the item has been revealed.

2. No individual-to-team bridge. Team outcome models either require team-level parameters that do not generalize to new compositions (team Elo, team-level IRT) or aggregate individual scores naively (max, mean). Regularized Adjusted Plus-Minus (RAPM) (Sill, 2010) in sports analytics isolates individual contributions via ridge regression but does not model latent synergy that transfers compositionally to unseen team configurations. Elo (Glickman and Jones, 1999; Pelánek, 2016) and TrueSkill (Herbrich et al., 2007) rank players or teams from match outcomes but do not decompose multidimensional skill or model within-item difficulty.

3. Disconnected assessment modes. No existing model connects individual assessment (tossup buzzes) to team prediction (bonus conversion) through shared latent parameters, despite the two being naturally linked in the same competition.

JBG: Redundant, just focus on contributions

We introduce CompIRT, a two-component probabilistic model that addresses all three limitations:

Task. We formalize *individual-to-team skill prediction*: given observations of individuals answering items of dynamic difficulty, predict the outcomes of novel team compositions on collaborative items. We use QB as the first benchmark (Section 2).

Model. *Pyramidal* IRT extends multidimensional IRT with position-dependent difficulty and a discrete-time survival buzz model. *Compositional* IRT aggregates individual latent skills with learnable pairwise synergy profiles to predict team outcomes. Individual skills are shared across both components (Section 3).

Evaluation. We design a stratified evaluation with temporal train/test splits, a four-level compositional generalization test on recombined teams, and within-tournament adaptation analysis measuring cold-start convergence, using nine tournaments spanning three years (Section 4).

Analysis. Interpretable skill profiles, synergy quantification, difficulty curves, and team composition recommendations (Section 5).

2 Background

2.1 Quiz Bowl

QB is a team knowledge competition governed by standardized rules (Eltinge, 2013). A match consists of two four-player teams answering a packet of approximately twenty *tossup* questions and up to twenty *bonus* questions per half.

Tossups. A moderator reads a tossup question aloud at a steady pace. The question is *pyramidal*: clues are arranged in descending order of obscurity, beginning with a difficult lead-in and ending with an easy "giveaway" clue (Rodriguez et al., 2019). At any point, a player from either team may *buzz in* and attempt to answer. If correct, the player's team earns 10 points[1] and gains access to a bonus question. If incorrect—a *neg*—the player's team loses 5 points and is locked out for the remainder of that tossup, while the opposing team hears the rest of the question. If no player buzzes by the end, the tossup is *dead* and no points are awarded.

The pyramidal structure makes the *position* of a buzz as informative as its correctness: a player who buzzes correctly after only two sentences demonstrates deeper knowledge than one who buzzes at the giveaway. This position–difficulty relationship is the key phenomenon that Pyr-IRT models (Section 3.1).

Bonuses. A bonus is awarded to the team that correctly answers the preceding tossup. Each bonus has three parts, each worth 10 points (maximum 30). The moderator directs the bonus to a specific player, but the *entire team confers* before answering. Parts typically share a common theme and may span a different category than the tossup that earned them.

Team conferral makes bonuses a natural site for studying composition effects: a team's score depends not just on individual knowledge but on communication, coverage of knowledge areas, and collaborative dynamics—the phenomena that Comp-IRT models (Section 3.2).

Tournament structure. ACF tournaments have a preliminary round-robin phase followed by elimination brackets. Teams are affiliated with schools or, at open tournaments such as the Chicago Open, formed ad hoc from players who may play on different school teams at other events. This natural variation in team composition across tournaments provides the cross-team player overlap needed to identify synergy parameters and to test compositional generalization (Section 4.3). Standard community metrics include points per game (PPG) for individuals and points per bonus (PPB) for teams.

2.2 Item Response Theory

IRT is a family of probabilistic models that relate a latent person ability θ to the probability of a correct response on a test item (van der Linden and Hambleton, 2013; Embretson and Reise, 2000).

Unidimensional IRT. The two-parameter logistic (2PL) model gives the probability that person i answers item j correctly as

$$P(y_{ij} = 1) = \sigma\big(a_j(\theta_i - b_j)\big), \qquad (1)$$

[1] Some tournaments award 15 points for correct buzzes before a designated "power mark" early in the question. We treat power buzzes as correct buzzes with a position indicator.

where $\theta_i \in \mathbb{R}$ is ability, $b_j \in \mathbb{R}$ is item difficulty, $a_j > 0$ is item discrimination, and σ is the logistic function. The key property of IRT is *separability*: person and item parameters are estimated jointly on a common scale, enabling fair comparison across different items and persons.

Multidimensional IRT. When items tap multiple skills, the compensatory MIRT model replaces the scalar ability with a skill vector $\boldsymbol{\theta}_i \in \mathbb{R}^K$ and the scalar discrimination with a vector $\boldsymbol{\alpha}_j \in \mathbb{R}^K$ (Reckase, 2009):

$$P(y_{ij} = 1) = \sigma\big(\boldsymbol{\alpha}_j^\top \boldsymbol{\theta}_i - d_j\big), \tag{2}$$

where d_j is a scalar difficulty intercept. High ability on one dimension can compensate for low ability on another.

Both models assign a *single, fixed* difficulty to each item. For pyramidal tossups, this is insufficient: the same question at different buzz positions is effectively a different item. Neither model composes individual abilities into a team-level prediction—the two limitations that COMPIRT addresses.

3 COMPIRT

COMPIRT has two components. PYR-IRT (Section 3.1) extends multidimensional IRT with position-dependent difficulty and a survival-process buzz model for tossups. COMP-IRT (Section 3.2) aggregates individual latent skills with learnable pairwise synergy profiles to predict team bonus outcomes. Individual skill parameters $\boldsymbol{\theta}$ are shared across both components: the tossup model estimates them; the bonus model consumes them (Section 3.3).

3.1 Pyramidal IRT

Let there be N players and J tossup questions, where question j consists of T_j tokens. For each player–question pair (i, j), we observe a tuple $(t^*_{ij}, y_{ij}, z_{ij})$: the buzz position t^*_{ij} (in tokens), the correctness indicator $y_{ij} \in \{0, 1\}$, and the observation type $z_{ij} \in \{\text{BUZZ}, \text{NO-BUZZ}, \text{CENSORED}\}$. A player who buzzes has $z_{ij} = \text{BUZZ}$; one who never buzzes on a dead tossup has $z_{ij} = \text{NO-BUZZ}$; one whose potential buzz is preempted by another player buzzing first has $z_{ij} = \text{CENSORED}$ at the observed buzz time t^*.

Latent parameters. Each player i has a *skill profile* $\boldsymbol{\theta}_i \in \mathbb{R}^K$ over K knowledge dimensions and a *confidence threshold* $c_i \in \mathbb{R}$ governing risk tolerance when deciding to buzz. Each question j has a *discrimination vector* $\boldsymbol{\alpha}_j \in \mathbb{R}^K$ encoding which skill dimensions the question loads on, and a *base difficulty* $\beta_j \in \mathbb{R}$ capturing the inherent obscurity of the answer.

Position-dependent difficulty. Pyramidal tossups arrange clues from obscure to obvious, so effective difficulty decreases as more tokens are revealed. We model the difficulty at token position t as

$$d_j(t) = \beta_j \cdot \sigma\big(\gamma_j(\mu_j - t/T_j)\big), \tag{3}$$

where σ is the logistic function, $\mu_j \in (0, 1)$ is the *inflection point* (the fractional position where difficulty transitions most rapidly), and $\gamma_j > 0$ controls the sharpness of the transition. This captures the empirical observation that well-written pyramidal questions have a "difficulty cliff" at the key clue, rather than a smooth monotonic decrease. Difficulty at the start of the question approximates β_j and at the end approaches zero. We compare exponential and linear decay alternatives in Appendix D.

Correctness model. Given that player i buzzes at position t on question j, the probability of a correct answer follows a compensatory multidimensional IRT model:

$$P(y_{ij} = 1 \mid t) = \sigma\big(\boldsymbol{\alpha}_j^\top \boldsymbol{\theta}_i - d_j(t)\big). \tag{4}$$

The inner product $\boldsymbol{\alpha}_j^\top \boldsymbol{\theta}_i$ measures the match between the player's skill profile and the question's knowledge requirements.

Buzz timing as a survival process. The decision of *when* to buzz is a discrete-time survival process. At each token position t, player i has a latent *hazard* of buzzing:

$$h_{ij}(t) = \sigma\big(\boldsymbol{\alpha}_j^\top \boldsymbol{\theta}_i - d_j(t) - c_i\big). \tag{5}$$

The hazard shares the same skill–difficulty interaction as the correctness model but includes an additional threshold c_i: a player buzzes when their estimated probability of being correct exceeds a personal confidence threshold $\sigma(c_i)$. Cautious players have large c_i; aggressive players have small c_i.

The *survival function*—the probability of not having buzzed before position t—is

$$S_{ij}(t) = \prod_{s=1}^{t-1} \big(1 - h_{ij}(s)\big). \tag{6}$$

Competing risks in game play. In a competitive game, multiple players listen simultaneously and only the first buzzer is observed. This is a *competing risks* model (Kalbfleisch and Prentice, 2002). Suppose player i^* buzzes at position t^*. The likelihood is

$$\mathcal{L}_{\text{tossup}} = h_{i^*j}(t^*) \cdot S_{i^*j}(t^*) \cdot \prod_{i \neq i^*} S_{ij}(t^*) \cdot P(y_{i^*j} \mid t^*)^{y_{i^*j}} \big(1 - P(y_{i^*j} \mid t^*)\big)^{1-y_{i^*j}}. \quad (7)$$

For dead tossups (no buzz), every player contributes a right-censored term $S_{ij}(T_j)$.

Neg handling. An incorrect buzz (neg) is especially informative: the player was confident enough to buzz but wrong. A neg at t^* contributes to both the buzz likelihood (hazard exceeded threshold) and the correctness likelihood ($y = 0$). After a neg, the negging team is locked out; the opposing team continues in a single-team survival model over $[t^* + 1, T_j]$ (Appendix A).

3.2 Compositional IRT

The tossup model estimates individual skill profiles $\boldsymbol{\theta}_i$. We now model how these compose to produce team-level bonus outcomes.

Consider a bonus with parts $k \in \{1, 2, 3\}$ answered by team $\mathcal{T} = \{p_1, \ldots, p_M\}$ (typically $M = 4$). Each part has difficulty β_k and discrimination $\boldsymbol{\alpha}_k \in \mathbb{R}^K$. We decompose the team's probability of answering part k correctly into *aggregation* (who on the team can answer this?) and *synergy* (how well does the team collaborate?).

Aggregation: individual knowledge. The team's aggregate knowledge for part k is a smooth maximum over member abilities:

$$A_k(\mathcal{T}) = \tau \cdot \log \sum_{m=1}^{M} \exp\Big(\frac{\boldsymbol{\alpha}_k^\top \boldsymbol{\theta}_{p_m}}{\tau}\Big), \quad (8)$$

where $\tau > 0$ is a temperature parameter. As $\tau \to 0$, this approaches $\max_m \boldsymbol{\alpha}_k^\top \boldsymbol{\theta}_{p_m}$; as $\tau \to \infty$, it approaches the mean. We also consider the *noisy-OR*:

$$P_{\text{indep}}(k, \mathcal{T}) = 1 - \prod_{m=1}^{M} \big(1 - \sigma(\boldsymbol{\alpha}_k^\top \boldsymbol{\theta}_{p_m} - \beta_k)\big), \quad (9)$$

which models each member as independently capable of knowing the answer; it is therefore a natural *no-synergy baseline*.

Synergy: team interaction. Beyond individual knowledge, team bonus conversion depends on communication, complementarity, and collaborative dynamics. We refer to these interaction effects as *synergy*. Each player has a latent *synergy profile* $\boldsymbol{\psi}_i \in \mathbb{R}^S$. The synergy contribution for team $\mathcal{T}$ is the sum of pairwise compatibility scores:

$$\Phi(\mathcal{T}) = \sum_{m < m'} \boldsymbol{\psi}_{p_m}^\top \boldsymbol{\psi}_{p_{m'}}. \quad (10)$$

A positive dot product indicates complementary interaction styles; a negative value indicates redundancy or friction. Because synergy profiles are properties of *individuals*, not teams, the model can predict synergy for any novel composition from its members' estimated profiles.

Outcome model. Combining aggregation and synergy:

$$P(y_k = 1 \mid \mathcal{T}) = \sigma\big(A_k(\mathcal{T}) + \Phi(\mathcal{T}) - \beta_k\big). \quad (11)$$

Identifiability of synergy profiles. The synergy profiles $\boldsymbol{\psi}_i$ are identifiable only if sufficient players appear on multiple distinct teams, providing the variation needed to separate individual synergy contributions from team-level confounds. Define the *team-composition graph* G where players are nodes and edges connect co-teammates. The profiles are identifiable up to rotation if G is connected and has sufficient edge diversity—analogous to the lineup variation condition for RAPM (Sill, 2010). A further concern is *assortative team formation*: if strong players consistently team with strong players, $\boldsymbol{\psi}_i$ may absorb team quality rather than genuine interaction effects. Sufficient roster variation across tournaments—particularly from the Chicago Open, where players form ad hoc teams—is needed to disentangle the two. We verify connectivity and degree distribution empirically in Section 4.1 and report graph statistics in Appendix C.

3.3 Training and Inference

The two components share skill parameters $\boldsymbol{\theta}_i$. We adopt a two-stage procedure:

1. **Tossup stage.** Fit PYR-IRT on all tossup data using NUTS MCMC[2] to obtain posterior distributions over $\boldsymbol{\theta}_i$, $\boldsymbol{\alpha}_j$, β_j, μ_j, γ_j, c_i.

2. **Bonus stage.** Fix $\boldsymbol{\theta}_i$ at posterior means from Stage 1 and fit COMP-IRT on bonus data via

[2]Implemented in NumPyro: https://num.pyro.ai/

stochastic variational inference to estimate ψ_i, $\boldsymbol{\alpha}_k$, β_k, and τ.

We place weakly informative Gaussian and half-Cauchy priors on all parameters (Appendix B). For players observed in training, the 2025 prior is the training posterior. For *new* players, a flat $\mathcal{N}(\mathbf{0}, \mathbf{I})$ prior ignores the fact that a debut at Chicago Open signals higher ability than a debut at Regionals. We therefore condition the population prior on tournament tier: $\boldsymbol{\theta}_i \sim \mathcal{N}(\boldsymbol{\mu}_{\text{tier}}, \Sigma_{\text{tier}})$, where $\boldsymbol{\mu}_{\text{tier}}$ and Σ_{tier} are estimated from training data. We ablate this against the flat prior in Section 5.3. As a further ablation, we consider end-to-end joint optimization via variational inference over the combined likelihood.

4 Experiment Design

4.1 Data

We use data from nine ACF and open-format QB tournaments spanning three years: **ACF Regionals** (2023–2025), the largest collegiate tournament by participation; **ACF Nationals** (2023–2025), the premier invitational; and the **Chicago Open** (2023–2025), where players form ad hoc teams, often with different partners than at their school events.

For each tournament, we collect: (1) tossup buzzpoint data—player identity, question identity, buzz position (in words), and correctness—from `quizbowlstats.com/buzzpoints`; (2) question text, answer lines, and categories from the QBReader API;[3] and (3) team rosters and game scores from tournament stat sheets.

Player entity linking. Players are linked across tournaments by `(first_name, last_name, school)` tuple with fuzzy matching for school name variants (e.g., "UMD" and "Maryland") and manual review of ambiguous cases. Details appear in Appendix F.

Team-composition graph. To verify that synergy profiles ψ_i are identifiable (Section 3.2), we construct the team-composition graph G: players are nodes and edges connect co-teammates. The Chicago Open tournaments are the primary source of cross-team variation, as players regularly form different teams than at their school tournaments. Graph statistics appear in Appendix C.

[3] https://www.qbreader.org/api

4.2 Train/Test Split

Primary: temporal split. We train on 2023 and 2024 tournaments (six tournaments) and evaluate on 2025 tournaments (three tournaments), reflecting the realistic setting of estimating skills from historical data and predicting outcomes at future events.

Question parameter estimation. Because test questions are from new packets unseen during training, question parameters ($\boldsymbol{\alpha}_j$, β_j, μ_j, γ_j) are estimated transductively on the test data alongside predictions—the standard psychometric practice of co-estimating items and persons. We also report *inductive* results where question parameters are predicted from metadata (category, answer line) without fitting question-specific parameters.

Secondary: leave-one-series-out. We hold out one tournament series (e.g., all three years of Chicago Open) and train on the remaining two, testing generalization across difficulty tiers.

4.3 Evaluation Scenarios

Scenario A: tossup prediction. We measure buzz position via MAE and Median Absolute Error on relative position $t^*/T_j \in [0, 1]$, correctness via AUC-ROC and log-loss, expected points per tossup under the model (capturing the risk–reward tradeoff), and calibration via ECE with reliability diagrams.

Scenario B: bonus prediction. We measure part-level AUC-ROC, log-loss, and accuracy for each of three binary predictions per bonus; bonus-level accuracy for predicting the exact score (0/10/20/30) and MAE of predicted PPB at the game level; and part-level calibration via ECE.

Scenario C: compositional generalization. We stratify test teams by training-time observability: **C1** (familiar teams, same roster as training); **C2** (recombined, all members seen but on different teams); **C3** (partially new, some members unseen); **C4** (fully new, no members seen). We report bonus metrics separately per group. The key test is **C2**: if CompIRT with synergy outperforms CompIRT without synergy on recombined teams, the synergy profiles generalize compositionally. Within **C3**, we further stratify by the fraction of unknown members (1-of-4 vs. 3-of-4) to measure how prediction degrades as known-player anchoring weakens. For **C4**, the bar is not accuracy but graceful degrada-

tion: predictions should revert to tier-level base rates, not collapse.

Scenario D: within-tournament adaptation. Independently of C1–C4, we measure online adaptation speed. For each player in the 2025 test set, we use the first k tossup observations from that tournament to update $\boldsymbol{\theta}_i$ (and $\boldsymbol{\psi}_i$ for teams) via posterior updating, then predict on remaining questions. Sweeping $k \in \{0, 5, 10, 20, \text{all}\}$ yields a learning curve: for returning players, this tests how quickly observation corrects the prior (a diagnostic of prior quality); for new players, it measures how many questions are needed before the model outperforms tier-level base rates.

Skill attribution validation and cross-difficulty transfer analysis (testing whether player rankings are stable across tournament tiers) appear in Appendices J and G.

4.4 Modeling Assumptions

We state four assumptions. **(1)** Buzz decisions are primarily skill-driven: the model treats the buzz point as a noisy threshold on $P(\text{correct} \mid \boldsymbol{\theta}_i, d_j(t))$ and does not condition on game state (score margin, remaining questions). **(2)** Bonus responses reflect team knowledge, not communication failure: if someone on the team knows the answer, the team gets credit. Communication breakdowns are absorbed into the synergy term, making "synergy" a composite of knowledge complementarity and communication effectiveness. **(3)** Synergy profiles are player-intrinsic and context-independent: player i has the same $\boldsymbol{\psi}_i$ regardless of teammates. This sacrifices higher-order interaction specificity for tractability and out-of-sample prediction. **(4)** Skill dimensionality K is sufficient to capture category-level variation but low enough to be identifiable from per-player data. We treat K as an empirical hyperparameter (Appendix E).

4.5 Baselines

Tossup. 2PL IRT (Equation 1, ignoring buzz position); Position-LR (logistic regression on category, t/T_j, $(t/T_j)^2$, PPG); Elo-Position (Pelánek, 2016) (Elo adapted so that earlier correct buzzes earn larger updates); PYR-IRT-1D (K=1 ablation); Neural Buzzer (MLP on player embedding, category, t/T_j).

Bonus. Noisy-OR (Equation 9); Best-member ($P = \max_m P_m$); Mean-member ($P = \text{mean}_m P_m$); Team Elo (team-level Elo on bonus outcomes); RAPM-style (Sill, 2010) (ridge regression with player indicators); COMP-IRT (no synergy, $\boldsymbol{\psi} = \mathbf{0}$); Neural Team (MLP on concatenated player embeddings).

All metrics include 95% bootstrap confidence intervals; pairwise comparisons use permutation tests ($p < 0.05$). Hyperparameter sweeps over K, S, difficulty curve, and aggregation function appear in Appendix E.

[ToDo: Plot: x-axis = number of observed tossups k, y-axis = prediction log-likelihood on remaining questions. Separate curves for returning players (prior = training posterior) and new players (prior = tier-conditioned population). Include a horizontal line for the tier-level base-rate predictor.]

Figure 1: Adaptation learning curves on 2025 tournaments. Returning players converge quickly, confirming prior quality; new players cross the base-rate threshold after $\sim k$ tossups. [ToDo: Fill after experiments.]

5 Results

5.1 Tossup Prediction (Scenario A)

Table 1 reports tossup prediction metrics.

[ToDo: Discuss: (1) PYR-IRT vs. baselines on correctness prediction; (2) value of multidimensionality ($K > 1$); (3) buzz position prediction quality; (4) calibration.]

5.2 Bonus Prediction (Scenario B)

Table 2 reports bonus prediction metrics.

[ToDo: Discuss: (1) COMP-IRT vs. baselines on part-level prediction; (2) value of synergy term; (3) aggregation function comparison (LSE vs. noisy-OR); (4) PPB prediction at the game level.]

5.3 Compositional Generalization (Scenario C)

Table 3 reports bonus AUC stratified by team familiarity.

[ToDo: Discuss: (1) AUC degradation from C1 to C4; (2) whether synergy improves C2 predictions; (3) comparison with RAPM, which cannot generalize to unseen lineups; (4) implications for team formation recommendations.]

5.4 Within-Tournament Adaptation (Scenario D)

Figure 1 shows prediction log-likelihood as a function of tossups observed within a tournament.

Model	Buzz MAE ↓	Buzz MedAE ↓	Correct AUC ↑	Correct LL ↓	ECE ↓
2PL IRT	–	–	–	–	–
Position-LR	–	–	–	–	–
Elo-Position	–	–	–	–	–
Neural Buzzer	–	–	–	–	–
PYR-IRT-1D	–	–	–	–	–
PYR-IRT (ours)	–	–	–	–	–

Table 1: Tossup prediction on 2025 test tournaments. Buzz position is relative ($t^*/T_j \in [0, 1]$). 95% bootstrap CIs appear in Appendix. [ToDo: Fill after experiments.]

Model	Part AUC ↑	Part LL ↓	PPB MAE ↓	ECE ↓
Best-member	–	–	–	–
Mean-member	–	–	–	–
Noisy-OR	–	–	–	–
Team Elo	–	–	–	–
RAPM-style	–	–	–	–
Neural Team	–	–	–	–
COMP-IRT (no synergy)	–	–	–	–
COMP-IRT (ours)	–	–	–	–

Table 2: Bonus prediction on 2025 test tournaments. Part-level metrics averaged across all three parts. [ToDo: Fill after experiments.]

Model	C1 Familiar	C2 Recomb.	C3 Partial	C4 New
Noisy-OR	–	–	–	–
RAPM-style	–	–	–	–
COMP-IRT (no syn.)	–	–	–	–
COMP-IRT (ours)	–	–	–	–

Table 3: Bonus AUC stratified by team familiarity. C2 (recombined teams) is the key test of compositional generalization. [ToDo: Fill after experiments.]

[ToDo: Discuss: (1) how many observations before new players outperform base rates; (2) whether returning player priors are well-calibrated (fast convergence = good prior); (3) comparison of tier-conditioned vs. flat prior for new players at $k = 0$.]

Ablation studies over K, S, difficulty curve family, and aggregation function appear in Appendix H. Qualitative analysis of difficulty curves, skill profiles, synergy heatmaps, and counterfactual team compositions appears in Appendix I.

6 Related Work

IRT in psychometrics and education. Item Response Theory underpins modern educational testing (van der Linden and Hambleton, 2013; Embretson and Reise, 2000). Multidimensional IRT (MIRT) extends the framework to items that tap multiple latent traits (Reckase, 2009). Deep-IRT (Yeung, 2019) and DEEPIRTOOLS (Urban and Bauer, 2021) combine neural networks with IRT structure for knowledge tracing; Deep Knowledge Tracing (Piech et al., 2015) models temporal skill evolution with recurrent networks. All assign a fixed difficulty to each item and model individuals, not teams. COMPIRT extends MIRT with position-dependent difficulty and a compositional team model.

IRT for NLP evaluation. IRT has been applied to NLP benchmark evaluation to estimate item difficulty and model ability (Lalor et al., 2019, 2024; Rodriguez et al., 2021; Vania et al., 2021). The `py-irt` library (Lalor et al., 2024) provides scalable estimation for standard models. These applications use *static* IRT on fixed-form benchmarks; COMPIRT addresses the distinct challenge of *dynamic* items whose difficulty changes as information is revealed.

Response-time IRT. Van der Linden's hierarchical framework (van der Linden, 2007) jointly models response accuracy and response time by positing separate first-level models linked through correlated latent traits (ability and speed) (Fox and Glas, 2001). A key distinction: response-time IRT treats speed as an *exogenous* latent trait measured by how quickly a person finishes an item. In PYR-IRT, the buzz time is an *endogenous decision*—the player chooses when to buzz based on their confidence—and the difficulty is a function of *how much* of the

item has been revealed, not of how fast the person works.

Crowdsourcing quality models. The GLAD model (Whitehill et al., 2009) and the Dawid-Skene model (Dawid and Skene, 1979) estimate latent annotator ability and item difficulty from noisy crowdsourced labels, with GLAD structurally equivalent to two-parameter IRT (Welinder et al., 2010). These models assume *independent* annotators on the same item; in bonus questions, team members *collaborate*, introducing interaction effects absent from crowdsourcing models. COMP-IRT generalizes crowdsourcing quality models from independent annotators to collaborating teams by adding synergy parameters.

Collective intelligence and team cognition. Woolley et al. (Woolley et al., 2010) show that human groups exhibit a collective intelligence factor (c-factor) that predicts team outcomes across diverse tasks and is not reducible to average member intelligence. Almaatouq et al. (Almaatouq et al., 2024) find that average individual skill is the strongest predictor of group outcomes but that social sensitivity and skill diversity provide additional signal. DeChurch and Mesmer-Magnus (DeChurch and Mesmer-Magnus, 2010) meta-analyze team cognition constructs. These results confirm that team outcomes involve interaction effects beyond individual abilities, but none of these studies provide a probabilistic model that decomposes team outcomes into individual and interaction components on a calibrated scale.

Sports analytics: individual-to-team models. Adjusted Plus-Minus (APM) (Rosenbaum, 2004) and its regularized variant (RAPM) (Sill, 2010) decompose team point differentials into individual player contributions via regression with player indicator variables. Pelechrinis and Papalexakis (Pelechrinis and Papalexakis, 2016) detect lineup-level synergy with tensor decomposition. RAPM solves a structurally analogous problem to COMP-IRT: estimating individual contributions to team outcomes. However, RAPM is a *discriminative* regression model that requires observed lineup data; it cannot predict outcomes for a team composition never seen during training. COMP-IRT is a *generative* probabilistic model with per-player synergy profiles, enabling out-of-sample team composition predictions—the core of our evaluation (Section 4.3, Scenario C2).

Quiz bowl AI systems. Rodriguez et al. (Rodriguez et al., 2019) frame QB as an incremental question answering task with three challenges: factoid QA, confidence calibration, and sequential decision-making. Earlier QB systems include QANTA (Iyyer et al., 2014) and adversarial question writing (Wallace et al., 2019). These systems build AI *players*; we build *psychometric models of human players*. The two are complementary: AI systems benefit from calibrated difficulty estimates (which PYR-IRT provides), and psychometric models can use AI outputs as a reference point.

Multi-agent LLM composition. Recent work composes multiple LLM agents into collaborative systems (Guo et al., 2024). LLM-Blender (Jiang et al., 2023) and Mixture-of-Agents (Wang et al., 2024) learn to combine outputs from heterogeneous models; FrugalGPT (Chen et al., 2023) routes queries to the cheapest capable model. These methods select or aggregate models based on task features, but lack a *latent ability model* that decomposes individual model strengths and predicts how compositions of models interact. COMPIRT's contribution is orthogonal: we provide a formal framework—shared latent skill vectors and synergy profiles—for the composition problem itself. The structural requirements (individual-level evaluation, variation in compositions, paired solo/group observations) are satisfiable in LLM evaluation settings where models are benchmarked individually and in ensembles (Rodriguez et al., 2021), making COMPIRT a candidate framework for principled multi-model team formation (Section 7).

7 Conclusion

We introduced COMPIRT, a probabilistic model that estimates individual skills from solo performance and composes them to predict team outcomes on collaborative tasks. *Pyramidal* IRT extends multidimensional IRT with position-dependent difficulty and a competing-risks survival model for incrementally revealed items. *Compositional* IRT aggregates individual latent skills with learnable pairwise synergy profiles, enabling out-of-sample predictions for novel team compositions.

[ToDo: Summarize key empirical findings after experiments: (1) tossup prediction quality; (2) value of synergy in bonus prediction; (3) compositional generalization results (Scenario C2); (4) within-tournament adaptation speed (Scenario D); (5) interpretability of skill profiles and difficulty

curves.]

Beyond quiz bowl, the framework applies wherever agents are assessed individually and then composed into teams: collaborative exams, clinical team diagnostics, organizational team formation, and LLM ensemble design. The two structural requirements are overlapping group membership (for synergy identifiability) and paired individual–group assessment on comparable items.

Future work: LLM agent composition. CompIRT's formalism extends to multi-model LLM systems where composition is typically handled by heuristics. Three concrete extensions follow. **(1)** *Model skill estimation*: run N LLMs on individual benchmarks with IRT-estimated item parameters (Rodriguez et al., 2021) to obtain per-model skill vectors $\boldsymbol{\theta}_i$ across capability dimensions (reasoning, knowledge, code, etc.). **(2)** *Ensemble synergy*: evaluate subsets of models on collaborative tasks (debate, multi-agent reasoning, ensemble voting) and fit Comp-IRT to estimate pairwise synergy profiles $\boldsymbol{\psi}_i$, identifying which model pairs amplify or cancel each other's strengths. **(3)** *Composition recommendation*: given a budget constraint and a task profile, use CompIRT to predict the optimal model team—analogous to our counterfactual team analysis, but for selecting which agents to deploy. These extensions require paired individual/ensemble evaluation data, which emerging multi-agent benchmarks are beginning to provide (Guo et al., 2024).

Limitations

Data. Our model relies on buzzpoint data and team rosters from tournament statistics websites. Some events lack word-level buzz positions, reducing the tossup signal to question-level correctness. Player entity linking across tournaments may miss valid links (conservative matching) or create spurious ones (aggressive matching), introducing noise in skill estimates.

Modeling assumptions. Section 4.4 states four assumptions explicitly; here we note their consequences. Ignoring game state in the buzz model (Assumption 1) means the model cannot capture strategic adjustments when a player is ahead or behind. Treating synergy as a composite of knowledge complementarity and communication (Assumption 2) means these effects cannot be disentangled without richer annotation. Context-independent synergy profiles (Assumption 3) sacrifice specificity—a player may collaborate differently with a dominant partner than a quiet one—for out-of-sample generalizability. Assortative team formation (strong players clustering together) may further confound $\boldsymbol{\psi}_i$ with team quality despite roster variation from the Chicago Open.

Scope. Our evaluation covers top-tier collegiate QB (ACF and Chicago Open). Generalization to high school tournaments, other difficulty tiers, or other team-knowledge domains requires further validation. The model does not incorporate question text features; text-conditioned item parameters are a natural extension but substantially expand scope.

Ethics Statement

Our data consists of publicly available tournament results and statistics. Player names appear in public records; we use them solely for entity linking and do not collect or infer any private information. Skill estimates and synergy profiles are intended as analytical tools for understanding team dynamics, not for evaluating individuals in high-stakes settings. We will anonymize player identities in all public-facing results and analyses.

Acknowledgments

[ToDo: Add acknowledgments.]

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A Competing Risks Likelihood Derivation

In a competitive tossup, eight players (four per team) simultaneously listen to the question. Let $h_{ij}(t)$ denote the hazard of player i buzzing on question j at token position t. The observed outcome is the identity i^* of the first buzzer and the position t^* at which they buzz. All other players' potential buzz times are right-censored at t^*.

The likelihood for a single tossup observation is:

$$\mathcal{L} = \underbrace{h_{i^*j}(t^*)}_{\text{buzzer's hazard}} \cdot \underbrace{\prod_{i \neq i^*} S_{ij}(t^*)}_{\substack{\text{others survive to } t^* \\ i \neq i^*}} \cdot \underbrace{S_{i^*j}(t^*)}_{\text{buzzer survived to } t^*} \cdot \underbrace{P(y_{i^*j} \mid t^*)^y (1 - P(y_{i^*j} \mid t^*))^{1-y}}_{\text{correctness likelihood}} \tag{12}$$

where $S_{ij}(t) = \prod_{s=1}^{t-1}(1 - h_{ij}(s))$ is the survival function.

For dead tossups (no player buzzes), all players contribute right-censored survival terms:

$$\mathcal{L}_{\text{dead}} = \prod_i S_{ij}(T_j) \tag{13}$$

When a player negs at t^*, their team is locked out for the remainder of the tossup. The opposing team then has the remaining question $[t^* + 1, T_j]$ to themselves, creating an interval-censored observation for the negging team and a modified competing risks model for the non-negging team.

[ToDo: Expand with the full neg-then-bounce-back derivation.]

B Prior Specification

We place weakly informative priors on all model parameters:

$$\begin{aligned} \boldsymbol{\theta}_i &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}_K) & \boldsymbol{\alpha}_j &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}_K) \\ \beta_j &\sim \mathcal{N}(0, 1) & c_i &\sim \mathcal{N}(0, 1) \\ \gamma_j &\sim \text{Half-Cauchy}(5) & \mu_j &\sim \text{Beta}(2, 2) \\ \boldsymbol{\psi}_i &\sim \mathcal{N}(\mathbf{0}, 0.5 \cdot \mathbf{I}_S) & \tau &\sim \text{Half-Cauchy}(1) \end{aligned} \tag{14}$$

Standard normal priors on $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$ center the ability and discrimination scales; the half-Cauchy prior on γ_j allows sharp difficulty transitions while regularizing against implausibly steep curves. The tighter prior on $\boldsymbol{\psi}$ encodes the assumption that synergy effects are secondary to individual skill.

For returning players at test time, the prior on $\boldsymbol{\theta}_i$ is the training-set posterior. For new players, we use a tier-conditioned population prior $\boldsymbol{\theta}_i \sim \mathcal{N}(\boldsymbol{\mu}_{\text{tier}}, \Sigma_{\text{tier}})$ where tier $\in$ {Regionals, Nationals, Chicago Open}, with $\boldsymbol{\mu}_{\text{tier}}$ and Σ_{tier} estimated from posterior skill distributions of training-set players in each tier.

C Data Statistics

D Difficulty Curve Comparison

[ToDo: Include comparison of exponential, linear, logistic, and non-parametric difficulty curves with held-out log-likelihood.]

E Hyperparameter Sensitivity

[ToDo: Include grid search results over skill dimensionality $K \in \{1, 3, 5, 10\}$, synergy dimensionality $S \in \{1, 2, 3, 5\}$, and aggregation function choice.]

F Player Entity Linking

[ToDo: Document the entity linking procedure, fuzzy matching thresholds, manual review protocol, and error analysis.]

G Cross-Difficulty Transfer

Players who appear at multiple 2025 tournament tiers (e.g., Regionals *and* Nationals) provide a natural test of whether skill estimates transfer across difficulty levels. If $\boldsymbol{\theta}_i$ is difficulty-invariant, a player's predicted rank relative to peers should be stable across tiers. If there is a difficulty $\times$ skill interaction—e.g., a player excels at Regionals-level questions but underperforms at Nationals-level ones—a scalar θ will fail to capture this, while the multidimensional $\boldsymbol{\theta}_i$ should absorb it into category-specific dimensions.

[ToDo: Discuss: (1) whether multidimensional skill captures cross-tier stability better than scalar; (2) which categories show the largest difficulty $\times$ skill interaction.]

H Ablation Studies

Table 7 isolates the contribution of each model component.

[ToDo: Discuss: (1) optimal K and S; (2) logistic vs. alternative difficulty curves; (3) LSE vs. noisy-OR aggregation; (4) two-stage vs. joint training.]

Tournament	Games	Teams	Players	Tossups	Buzzes	Negs	Bonuses
ACF Regionals 2023	–	–	–	–	–	–	–
ACF Regionals 2024	–	–	–	–	–	–	–
ACF Regionals 2025	–	–	–	–	–	–	–
ACF Nationals 2023	–	–	–	–	–	–	–
ACF Nationals 2024	–	–	–	–	–	–	–
ACF Nationals 2025	–	–	–	–	–	–	–
Chicago Open 2023	–	–	–	–	–	–	–
Chicago Open 2024	–	–	–	–	–	–	–
Chicago Open 2025	–	–	–	–	–	–	–
Total	–	–	–	–	–	–	–

Table 4: Per-tournament data statistics. [ToDo: Fill after data collection.]

Cross-tournament statistic	Value
Players in 2+ tournaments	–
Players on 2+ distinct teams	–
Connected components in team-composition graph	–
Avg. degree in team-composition graph	–

Table 5: Cross-tournament player overlap statistics for synergy identifiability analysis. [ToDo: Fill after entity linking.]

Model	Spearman $\rho \uparrow$	Rank MAE $\downarrow$
Elo	–	–
PYR-IRT-1D	–	–
PYR-IRT (ours)	–	–

Table 6: Rank correlation of predicted player performance across tournament tiers, restricted to players appearing at two or more 2025 tiers. [ToDo: Fill after experiments.]

Variant	Tossup AUC	Bonus AUC
Skill dimensionality		
$K = 1$	–	–
$K = 3$	–	–
$K = 5$	–	–
$K = 10$	–	–
Synergy dimensionality		
$S = 0$ (no synergy)	–	–
$S = 1$	–	–
$S = 3$	–	–
$S = 5$	–	–
Difficulty curve		
Exponential	–	–
Linear	–	–
Logistic (default)	–	–
Aggregation function		
LSE (default)	–	–
Noisy-OR	–	–
Hard max	–	–
Training procedure		
Two-stage (default)	–	–
End-to-end joint	–	–

Table 7: Ablation study over model components. [ToDo: Fill after experiments.]

I Qualitative Analysis

Difficulty curves. [ToDo: Plot estimated $d_j(t)$ for 5–10 representative questions alongside empirical buzz density histograms. Show that the logistic transition captures difficulty cliffs. Include a figure.]

Skill profiles. [ToDo: Show radar charts of multidimensional skill profiles for 5–10 well-known players, annotated with their known category strengths. Include a figure.]

Team synergy heatmap. [ToDo: For selected teams, visualize pairwise synergy scores $\boldsymbol{\psi}_{p_m}^\top \boldsymbol{\psi}_{p_{m'}}$ as a heatmap. Include a figure.]

Counterfactual team composition. [ToDo: Model-based "what-if" analysis: swap one player between two teams and predict the change in PPB. Discuss implications for team formation strategy.]

J Skill Attribution

Skill ranking validation. [ToDo: Report Spearman correlation between estimated $\boldsymbol{\theta}_i$ and community rankings.]

Category alignment. [ToDo: Report cosine similarity between learned $\boldsymbol{\alpha}_j$ vectors and ground-truth category indicators. Visualize the top-3 skill dimensions and the categories they load on.]

Synergy decomposition. [ToDo: For representative teams, decompose predicted bonus outcome into individual skill contribution and synergy contribution. Show that synergy correlates with PPB residuals.]